Housing Price Modeling in R
Technical Summary Report
Cristina McMillin

1. Introduction

This analysis examines relationships between housing characteristics and sale prices using a dataset containing 2,692 homes and 23 variables. Three regression models are evaluated: a multiple regression model containing quantitative and categorical predictors, a complete second-order model containing linear, squared, and interaction terms, and a reduced model without the squared terms.

Exploratory scatterplots and Pearson correlation coefficients are used to examine relationships between selected variables. Each model is evaluated using its coefficients, coefficient of determination, significance tests, residual diagnostics, and prediction results. Finally, a nested-model F-test determines whether adding the squared terms significantly improves the model based on school ratings and crime rates. If the models prove to be significant, they could help real estate companies set better prices when listing a home for a client and ensure that listings can be sold within a reasonable amount of time.

The variables I will be working with are:

- "price" – sale price of the home;
- "bathrooms" – number of bathrooms;
- "sqft_living" – size of the living area in sqft;
- "sqft_above" – size of the upper level in sqft;
- "age" – age of the home;
- "crime" – crime rate per 100,000 people;
- "school_rating" – average rating of schools in the area;
- "view" – qualitative variable – home backs out to a lake (view=2), backs out to trees (view=1), or backs out to a road (view=0)

2. Model #1 - First Order Regression Model with Quantitative and Qualitative Variables

Correlation Analysis

Before I create the multiple regression model, I will analyze the relationship between price and living area and between price and the age of a home by creating two scatterplots. After this I will be analyzing the Pearson Correlation Coefficients of these variables.



	price	sqft_living	age
price	1.0000	0.6895	-0.0746

The scatterplot of price vs. living area shows a positive, approximately linear relationship between living area and price. The Pearson correlation coefficient of **0.6895** confirms a moderately strong positive association: larger living areas tend to be associated with higher sale prices.

The scatterplot of price vs. age does not show a strong marginal relationship between age and price. The Pearson correlation coefficient is **–0.0746**, indicating a very weak negative linear association. However, this bivariate correlation does not determine whether age will be significant after the other predictors are held constant in a multiple regression model.

Reporting Results

The general form of a first order regression model for price as the response variable and living area (sqft_living), upper-level area (sqft_above), age of the home, number of bathrooms, and view as predictor variables:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4 + \beta_5 X_5 + \beta_6 X_6 + \varepsilon$$

where Y– price, $\mathbf{X_1}$ – living area, $\mathbf{X_2}$ – upper-level area, X_3 – age, X_4 – number of bathrooms, X_5 – trees view (view1), X_6 – lake view (view2)

The prediction regression model is:

$$\widehat{Y} = b_0 + b_1 X_1 + b_2 X_2 + b_3 X_3 + b_4 X_4 + b_5 X_5 + b_6 X_6 + \varepsilon$$

where $\widehat{Y}$– predicted price, $\mathbf{X_1}$ – living area, $\mathbf{X_2}$ – upper-level area, X_3 – age, X_4 – number of bathrooms, X_5 – trees view (view1), X_6 – lake view (view2)

The output from R regarding the first order regression model:

```
Call:
lm(formula = price ~ sqft_living + sqft_above + age + bathrooms +
    view, data = housing)

Residuals:
    Min      1Q  Median      3Q     Max
-419299 -101792   -5606   93896  489323

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 7.709e+03  1.411e+04   0.546  0.58495
sqft_living 1.293e+02  8.123e+00  15.916  < 2e-16 ***
sqft_above  1.951e+01  7.458e+00   2.616  0.00894 **
age         1.451e+03  1.199e+02  12.098  < 2e-16 ***
bathrooms   4.397e+04  6.126e+03   7.178 9.13e-13 ***
view1       1.675e+05  1.071e+04  15.640  < 2e-16 ***
view2       2.490e+05  1.201e+04  20.739  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 133600 on 2685 degrees of freedom
Multiple R-squared:  0.6029,    Adjusted R-squared:  0.602
F-statistic: 679.3 on 6 and 2685 DF,  p-value: < 2.2e-16
```

From this output, the prediction model equation is:

$$\widehat{Y} = 7709 + 129.3X_1 + 19.51X_2 + 1451X_3 + 43,970X_4 + 167,500X_5 + 249,000X_6$$

where $\widehat{Y}$– predicted price, $\mathbf{X_1}$ – living area, $\mathbf{X_2}$ – upper-level area, X_3 – age, X_4 – number of bathrooms, X_5 – trees view (view1), X_6 – lake view (view2)

Model fit.

The model has an R^2 of **0.6029** and an adjusted R^2 of **0.6020**. Therefore, the predictors collectively explain approximately **60.3%** of the observed variation in sale prices. The similarity between the two values indicates that the adjustment for the number of predictors is small. Model adequacy must also be evaluated using residual diagnostics and predictive performance rather than an arbitrary R^2 threshold.

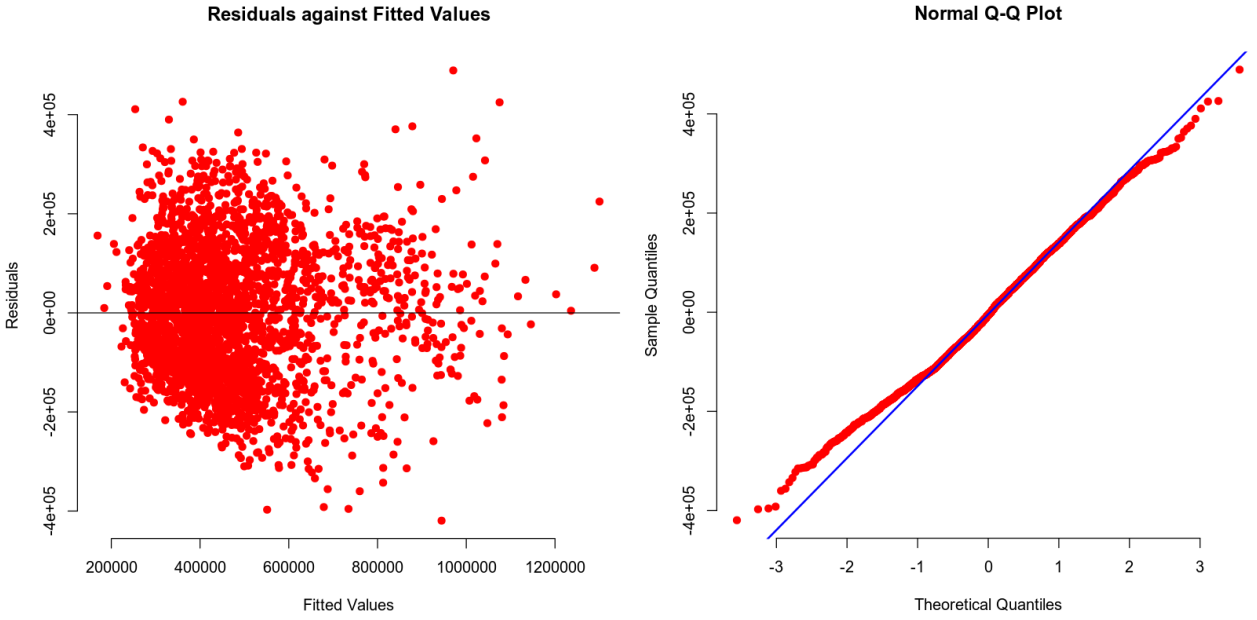
Beta estimates.

Holding the other predictors constant:

- each additional square foot of living area is associated with an estimated $129.30 increase in sale price;
- a home with a lake view is predicted to cost approximately $249,000 more than a comparable home with a road view, which is the reference category.

Residuals vs. Fitted Values.

Below are the following plots: "**Residuals against Fitted Values**" and "**Normal Q-Q plot**".



The residuals-versus-fitted plot shows greater dispersion at higher fitted values, along with several clusters and potential outliers. This pattern indicates heteroscedasticity, meaning that the

residual variance is not constant across the fitted values. Consequently, the usual standard errors, significance tests, and prediction intervals should be interpreted cautiously.

The normal Q–Q plot follows the reference line through much of its middle portion but departs from the line in both tails. This suggests that the residual distribution is not fully normal, particularly for extreme observations.

Evaluating Significance of Model

Overall F-test.

The null hypothesis is that no relationship exists between any of the predictor variables and the response variable – house price (Berrier et al., 2016).

$$H_0: \beta_1 = \beta_2 = \beta_3 = \beta_4 = \beta_5 = \beta_6 = 0$$

The alternative hypothesis is that a relationship exists with at least one variable.

$$H_a: \text{ At least one } \beta_i \neq 0 \text{ for } i = 1, 2, 3, 4, 5, 6$$

The overall **F**-statistic is **F = 679.3** with a corresponding **p**-value of $< 2.2 \times 10^{-16}$. Since the **p**-value is way smaller than the significance level $\alpha = 0.05$, we can reject the null hypothesis, which would suggest that the model is statistically significant. At least one predictor is significantly associated with sale price.

Individual t-tests:

For each predictor, the individual t-test evaluates $H_o: \beta = 0$ against $H_a: \beta \neq 0$. A term is considered statistically significant when its p-value is less than **0.05**.

PREDICTOR	ESTIMATE	T-STATISTIC	P-VALUE	RESULT
LIVING AREA	129.30	15.916	$< 2\times10^{-16}$	Significant
UPPER-LEVEL AREA	19.51	2.616	0.00894	Significant
AGE	1,451	12.098	$< 2\times10^{-16}$	Significant
BATHROOMS	43,970	7.178	9.13×10^{-13}	Significant
TREES VIEW	167,500	15.640	$< 2\times10^{-16}$	Significant
LAKE VIEW	249,000	20.739	$< 2\times10^{-16}$	Significant

At the 5% significance level, all six terms are statistically significant after the remaining predictors are held constant. The positive beta coefficient for age differs from its weak negative Pearson correlation with price because the regression coefficient measures the conditional relationship after controlling for living area, upper-level area, bathrooms, and view. This difference does not make the age coefficient invalid, although additional investigation would be needed to explain the relationship.

Making Predictions Using Model

Now, we will look at how well the model predicts a home's sale price based on its characteristics. Two types of intervals are reported. A confidence interval estimates the average sale price for homes with the specified characteristics. A prediction interval estimates the likely price range for one individual home with those characteristics. Prediction intervals are wider because individual homes vary in ways that the model does not fully capture, even when their measured characteristics are similar.

[1] "confidence interval"

A matrix: 1 × 3 of type dbl

fit	lwr	upr
459828.2	446087.9	473568.5

[1] "prediction interval"

A matrix: 1 × 3 of type dbl

fit	lwr	upr
459828.2	239563	680093.4

For the first prediction, we consider a home with 2,150 square feet of living area and 1,050 square feet of upper-level living area. The home is 15 years old, has three bathrooms, and backs onto a road. Its predicted price is **$459,828**. The 90% prediction interval for an individual property with these characteristics is **$239,563** to **$680,093**, while the 90% confidence interval for the mean price of comparable properties is **$446,088** to **$473,569**.

[1] "confidence interval"

A matrix: 1 × 3 of type dbl

fit	lwr	upr
1074285	1045117	1103454

[1] "prediction interval"

A matrix: 1 × 3 of type dbl

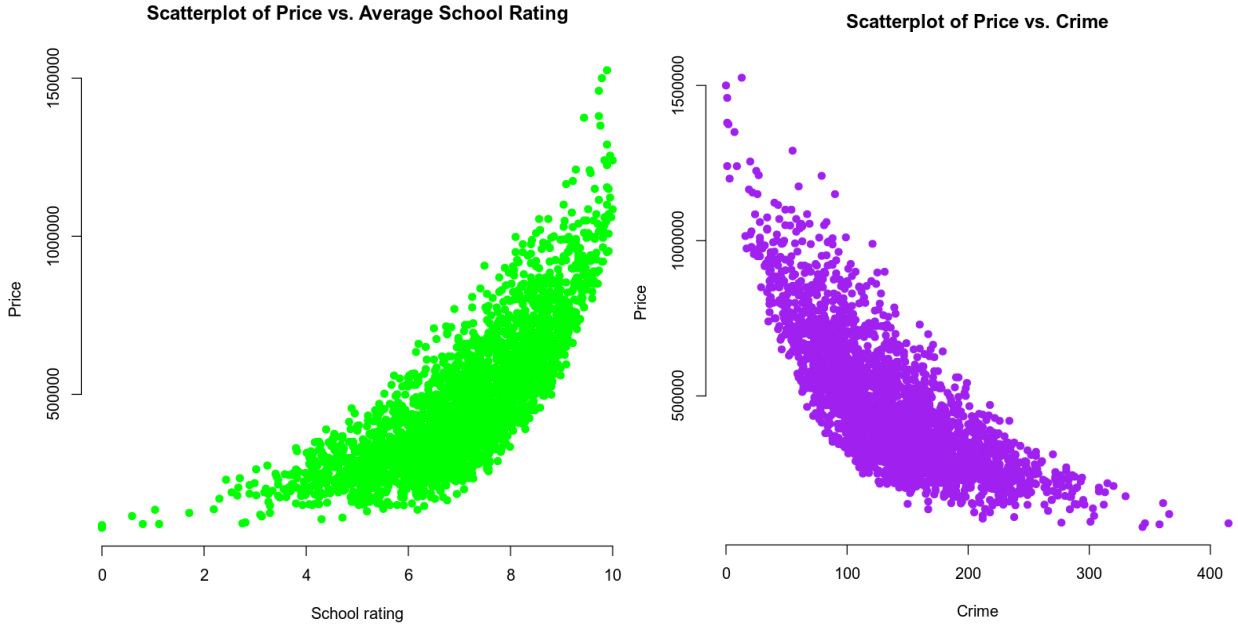
fit	lwr	upr
1074285	852522.6	1296048

For the second prediction, we consider a home with 4,250 square feet of living area and 2,100 square feet of upper-level living area. The home is five years old, has five bathrooms, and backs onto a lake. Its predicted price is **$1,074,285**. The 90% prediction interval is **$852,523** to **$1,296,048**, while the 90% confidence interval for the mean price of comparable properties is **$1,045,117** to **$1,103,454**.

The second property receives a substantially higher prediction because it is larger, has more bathrooms, and has a lake view.

3. Model #2 - Complete Second Order Regression Model with Quantitative Variables

Correlation Analysis



The relationship between school rating and price is positive but nonlinear. Prices increase relatively slowly across lower and middle school ratings and then rise more rapidly at the highest ratings, suggesting curvature that may not be captured adequately by a purely linear term.

The relationship between crime rate and price is negative and nonlinear across most of the observed range. Prices decline sharply at lower crime rates, after which the rate of decline becomes less pronounced.

Because both relationships display curvature, a second-order regression model containing linear, squared, and interaction terms is evaluated.

Reporting Results

The general form of a complete second order regression model for price as the response variable and average school rating in the area and crime rate per 100,000 people as predictor variables:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1 X_2 + \beta_4 {X_1}^2 + \beta_5 {X_2}^2 + \varepsilon,$$

where Y– price, $\mathbf{X_1}$ – school rating, $\mathbf{X_2}$ – crime

The prediction regression model is:

$$\hat{\mathbf{Y}} = \mathbf{b_0} + b_1 X_1 + b_2 X_2 + b_3 X_1 X_2 + b_4 {X_1}^2 + b_5 {X_2}^2$$

where $\hat{Y}$– predicted price, $\mathbf{X_1}$ – school rating, $\mathbf{X_2}$ – crime

The output from R regarding the complete second order regression model:

```
Call:
lm(formula = price ~ school_rating + crime + school_rating:crime +
    I(school_rating^2) + I(crime^2), data = housing)

Residuals:
    Min      1Q  Median      3Q     Max
-340729  -61055   -6288   56875  427915

Coefficients:
                     Estimate Std. Error t value Pr(>|t|)
(Intercept)          7.339e+05  1.032e+05   7.113 1.45e-12 ***
school_rating       -7.375e+04  2.083e+04  -3.541 0.000406 ***
crime               -3.155e+03  5.235e+02  -6.027 1.90e-09 ***
I(school_rating^2)   1.165e+04  1.109e+03  10.497  < 2e-16 ***
I(crime^2)           6.377e+00  7.265e-01   8.777  < 2e-16 ***
school_rating:crime -5.227e+01  4.853e+01  -1.077 0.281513
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 92690 on 2686 degrees of freedom
Multiple R-squared:  0.8088,    Adjusted R-squared:  0.8084
F-statistic:  2272 on 5 and 2686 DF,  p-value: < 2.2e-16
```

From this output, the prediction model equation is:

$$\hat{\mathbf{Y}} = \mathbf{733,900} - \mathbf{73,750} X_1 - \mathbf{3,155} X_2 - \mathbf{52.27} X_1 X_2 + \mathbf{11,650} {X_1}^2 + \mathbf{6.377} {X_2}^2$$

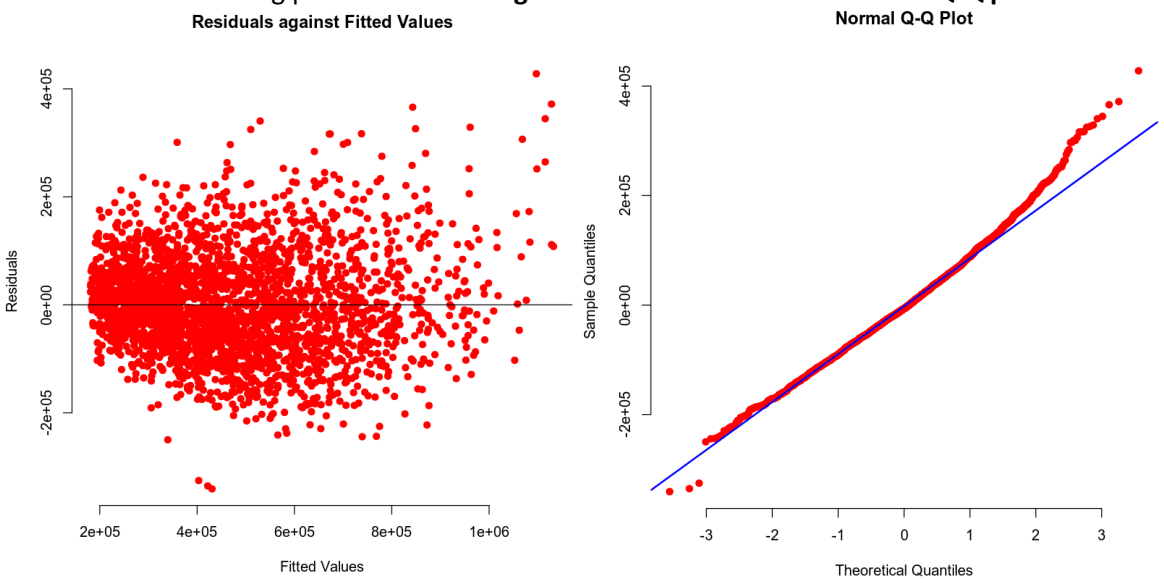
where $\hat{Y}$– predicted price, $\mathbf{X_1}$ – school rating, $\mathbf{X_2}$ – crime

Model fit.

The complete second-order model has an R^2 of **0.8088** and an adjusted R^2 of **0.8084**. The model therefore explains approximately **80.9%** of the observed variation in sale prices. The close agreement between the two values indicates that the penalty for including the additional terms is small.

Residuals vs. Fitted Values.

Below are the following plots: "**Residuals against Fitted Values**" and "**Normal Q-Q plot**".



The residuals-versus-fitted plot shows increasing dispersion at higher fitted values, indicating that the constant-variance assumption is not fully satisfied. Several clusters and potential outliers are also visible.

The normal Q–Q plot follows the reference line through most of its range but departs from it in the upper tail. The residual distribution is therefore approximately normal through its center but displays some non-normality among the largest residuals.

Evaluating Significance of Model

Overall F-test.

The null hypothesis is that no relationship exists between any of the predictor variables and the response variable – house price (Berrier et al., 2016).

$$H_0: \beta_1 = \beta_2 = \beta_3 = \beta_4 = \beta_5 = 0$$

The alternative hypothesis is that a relationship exists with at least one variable.

$$H_a: \text{ At least one } \beta_i \neq 0 \text{ for } i = 1, 2, 3, 4, 5$$

The overall **F**-statistic is **F = 2272** with a corresponding **p**-value of $< 2.2 \times 10^{-16}$. Since the **p**-value is way smaller than the significance level $\alpha = 0.05$, we can reject the null hypothesis, which would suggest that the model is statistically significant.

Individual t-tests:

TERM	ESTIMATE	T-STATISTIC	P-VALUE	RESULT
SCHOOL RATING	−73,750	−3.541	0.000406	Significant
CRIME	−3,155	−6.027	1.90×10^{-9}	Significant
SCHOOL RATING²	11,650	10.497	$< 2\times10^{-16}$	Significant
CRIME²	6.377	8.777	$< 2\times10^{-16}$	Significant
SCHOOL RATING × CRIME	−52.27	−1.077	0.281513	Not significant

At the 5% significance level, both linear terms and both squared terms are significant, while the interaction term is not. Because the model contains squared and interaction terms, the coefficients cannot be interpreted as independent, constant effects. Instead, the estimated relationship between each predictor and price varies across the values of both predictors. The significant squared terms support the curved relationships observed in the exploratory plots.

Making Predictions Using Model

Next, we will examine predictions from the complete second-order model based on school rating and crime rate. As in the previous analysis, the prediction interval estimates the likely price range for one individual home, while the confidence interval estimates the average price of comparable homes.

```
[1] "prediction interval"
```

A matrix: 1 × 3 of type dbl

fit	lwr	upr
874497	721606.2	1027388

```
[1] "confidence interval"
```

A matrix: 1 × 3 of type dbl

fit	lwr	upr
874497	863681.4	885312.7

For the first prediction, we consider a home in an area with an average school rating of 9.80 and a crime rate of 81.02 per 100,000 people. Its predicted price is **$874,497**. The 90% prediction interval for an individual property in an area with these characteristics is **$721,606 to $1,027,388**, while the 90% confidence interval for the average price of comparable properties is **$863,681 to $885,313**.

```
[1] "confidence interval"
```

A matrix: 1 × 3 of type dbl

fit	lwr	upr
199706.7	191753.5	207659.9

```
[1] "prediction interval"
```

A matrix: 1 × 3 of type dbl

fit	lwr	upr
199706.7	46991.65	352421.7

For the second prediction, we consider a home in an area with an average school rating of 4.28 and a crime rate of 215.50 per 100,000 people. Its predicted price is **$199,707**. The 90% prediction interval is **$46,992 to $352,422**, while the 90% confidence interval for the average price of comparable properties is **$191,754 to $207,660**.

Within this model, the combination of a higher school rating and lower crime rate produces a substantially higher predicted price. These results represent statistical associations and do not establish that school ratings or crime rates directly cause differences in home prices.

4. Nested Models F-Test

Reporting Results

The general form of a first order regression model for price as the response variable and average school rating in the area and crime rate per 100,000 people as predictor variables with an interaction term:

$$\boldsymbol{Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1 X_2 + \varepsilon,}$$

where $\boldsymbol{Y}$– price, $\mathbf{X_1}$ – school rating, $\mathbf{X_2}$ – crime

The prediction regression model is:

$$\widehat{\mathbf{Y}} = \mathbf{b_0} + \boldsymbol{b_1 X_1 + b_2 X_2 + b_3 X_1 X_2}$$

where $\widehat{\boldsymbol{Y}}$– predicted price, $\mathbf{X_1}$ – school rating, $\mathbf{X_2}$ – crime

The output from R regarding the first order regression model:

```
Call:
lm(formula = price ~ school_rating + crime + school_rating:crime,
    data = housing)

Residuals:
    Min      1Q  Median      3Q     Max
-336984  -63754   -4397   58894  440377

Coefficients:
                     Estimate Std. Error t value Pr(>|t|)
(Intercept)         -410233.37   25261.25  -16.24   <2e-16 ***
school_rating        155559.97    3133.06   49.65   <2e-16 ***
crime                  2230.07     129.70   17.20   <2e-16 ***
school_rating:crime    -564.85      17.86  -31.63   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 94870 on 2688 degrees of freedom
Multiple R-squared:  0.7995,    Adjusted R-squared:  0.7993
F-statistic:  3573 on 3 and 2688 DF,  p-value: < 2.2e-16
```

From this output, the prediction model equation is:

$$\widehat{\mathbf{Y}} = \mathbf{-410,233.37 + 155,559.97 X_1 + 2,230.07 X_2 - 564.85 X_1 X_2}$$

where $\widehat{\boldsymbol{Y}}$– predicted price, $\mathbf{X_1}$ – school rating, $\mathbf{X_2}$ – crime

Evaluating Significance of Model

Overall F-test.

The null hypothesis is that no relationship exists between any of the predictor variables and the response variable – house price (Berrier et al., 2016).

$$H_0: \beta_1 = \beta_2 = \beta_3 = 0$$

The alternative hypothesis is that a relationship exists with at least one variable.

$$H_a: \text{ At least one } \beta_i \neq 0 \text{ for } i = 1, 2, 3$$

The overall **F**-statistic is **F = 3573** with a corresponding **p**-value of $< 2.2 \times 10^{-16}$. Since the **p**-value is way smaller than the significance level $\alpha = 0.05$, we can reject the null hypothesis, which would suggest that the model is statistically significant.

Individual t-tests:

TERM	ESTIMATE	T-STATISTIC	P-VALUE	RESULT
SCHOOL RATING	155,559.97	49.65	$< 2\times10^{-16}$	Significant
CRIME	2,230.07	17.20	$< 2\times10^{-16}$	Significant
SCHOOL RATING × CRIME	−564.85	−31.63	$< 2\times10^{-16}$	Significant

Therefore, using the 5% significance level, all the terms are statistically significant. Looking at the sign of the beta coefficients though, not all the changes in price make sense. However, the model contains interaction, so the coefficients for school rating and crime cannot be interpreted independently. The estimated relationship between crime and price changes with school rating, while the estimated relationship between school rating and price changes with crime. Therefore, the positive coefficient for crime does not necessarily imply that higher crime predicts higher prices across the full dataset.

Model Comparison

I will now use the nested models F-test to compare this first order regression model with the complete second regression model we created and analyzed earlier. In general, a complete model includes all the squared terms and interaction between variables. The term 'reduced model' refers to any model that has been simplified by omitting certain interaction effects between variables or any squared terms of the variables included. In our case our reduced model includes the interaction between the predictor variables, but not their squared terms.

The general form and prediction equation of the model that is the <u>reduced</u> model is:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1 X_2 + \varepsilon \quad \textbf{general}$$

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X_1 + \hat{\beta}_2 X_2 + \hat{\beta}_3 X_1 X_2 \quad \textbf{predicted}$$

where Y– price, $\hat{Y}$– predicted price, X_1 – school rating, X_2 – crime.

The general form and prediction equation of the model that is the <u>complete</u> model is:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1 X_2 + \beta_4 {X_1}^2 + \beta_5 {X_2}^2 + \varepsilon \quad \textbf{general}$$

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X_1 + \hat{\beta}_2 X_2 + \hat{\beta}_3 X_1 X_2 + \hat{\beta}_4 {X_1}^2 + \hat{\beta}_5 {X_2}^2 \quad \textbf{predicted}$$

where Y– price, $\hat{Y}$– predicted price, X_1 – school rating, X_2 – crime.

The output from R regarding the nested models F-test:

Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
2686	2.307469e+13	NA	NA	NA	NA
2688	2.419501e+13	-2	-1.120319e+12	65.20513	2.22716e-28

Null hypothesis – the null hypothesis is that the beta coefficients for the squared terms are equal to zero:

$$\boldsymbol{H_0: \beta_4 = \beta_5 = 0}$$

Alternative hypothesis – the alternative hypothesis is that at least one of the beta coefficients for the squared terms are not equal:

$$\boldsymbol{H_a}: \text{ At least one } \boldsymbol{\beta_i \neq 0} \text{ for } \boldsymbol{i = 4, 5}$$

The nested models F = **65.20513** with a corresponding **p**-value of $\boldsymbol{2.22716 \times 10^{-28}}$**.** Therefore, the null hypothesis that both squared-term coefficients equal zero is rejected. The squared terms provide a statistically significant improvement in fit beyond the linear and interaction terms, supporting selection of the complete second-order model over the reduced model.

5. Conclusion

The analysis showed that housing prices have meaningful relationships with both property characteristics and neighborhood conditions. The first model, which used living area, upper-level area, age, bathrooms, and view, explained approximately 60% of the observed differences in prices. The model was statistically significant, but its diagnostic plots showed that its predictions were less consistent for higher-priced homes.

The models based on school ratings and crime rates explained approximately 80% of the variation in prices. The complete second-order model performed better than the reduced model because it accounted for the curved relationships observed in the data. The nested-model comparison confirmed that adding the squared terms significantly improved the model. Therefore, the complete second-order model provided the best fit among the three models evaluated.

However, this model should not yet be used as a complete pricing system. In reality, it is not possible to accurately predict a home's price based solely on the quality of its neighborhood's schools and crime statistics, as numerous other factors play significant roles in determining home values. Many factors contribute significantly to a home's desirability including the total square footage of the house and the accompanying yard space, the number of bedrooms and bathrooms available, and finally the overall quality of the home's construction. Future analysis should incorporate additional property and location characteristics and evaluate how accurately the model predicts new observations.

6. References

Berrier, H., Chan, C., & Pardoe, I. (2016). Applied Statistics with Data Analytics (R). zyBooks.com.